PRODUCT & TECHNICAL DOCUMENTATION

MillTrack

One digital record for a spinning mill: from the truck at the gate, through the godown, mixing, production and packing, to the vehicle that carries the yarn away.

Version 1.0 | Mobile application (Android, built with Expo / React Native)

Audience: mill owners and management, department heads, supervisors and operators, and the engineers who maintain the system.

How to read it: Part 1 explains the system in plain language. Part 2 is the technical reference.

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Login, Gate (vehicles, new entry, party picker, detail, settings), Godown (queue, unload setup, weighing, stock, mixing sheet, new lot), Supervisor (O/E, Card, Draw Frame, DOFF, lock, change shift), Packaging (pack bag, dispatch, dispatch detail, loose stock, paper log, finalize), Admin (overview, reports, users, register, analytics, pallis, trace, recipe).

1. Why MillTrack exists

A spinning mill turns raw cotton bales into yarn cones. Between the truck arriving and the yarn leaving, the material changes form many times and passes through many hands. Traditionally every step is written in a paper register. MillTrack replaces those registers with one mobile application that every department uses, so there is a single, trusted record of what came in, where it is, what was made and what went out.

1.1 The problems it solves

Problem with paper registers	What MillTrack does instead
Numbers are written late, copied, or lost. Nobody can say what is in the godown right now.	Every weight is entered once, at the moment it happens, on the phone of the person who did the work. Stock is always current.
It is hard to trace a bale of cotton to the yarn that was made from it.	Every bale gets a unique QR code at unloading. A bale can be followed into a mixing lot, and the Admin can trace any code.
Management waits days for a production or stock summary.	The Admin dashboard shows stock in the godown, in mixing, packed and dispatched, live, and can compare any two days.
Mistakes are corrected by overwriting, so the history is lost.	Locked shifts accept only visible corrections with a reason. Finished sheets are frozen into permanent PDF files that can never change.
Different people enter the same thing differently ("Super", "super spinners", "SS").	Clients, materials, counts, colours and cone types are chosen from lists maintained by the Admin, so names are consistent.

1.2 The intent, in one sentence

> **INTENT**
> Give every department a fast, simple screen that matches the paper register it replaces, and give management one dependable view of the whole mill, without adding work for the people on the floor.

1.3 Design principles

- **Looks like the paper sheet.** The Supervisor, Godown and Packaging sheets are grids that mirror the registers people already know.
- **Few taps, big targets.** Screens are built for one-handed use on the floor: large buttons, number pads, defaults that remember the last choice.
- **The server is the authority.** The phone never decides what shift it is, what a bag weighs or what stock exists. It only records and displays what the server says.
- **No silent failures.** Every error is shown in plain words (red message or banner). Mistakes can be corrected, but never erased.
- **One look.** One colour scheme, one set of buttons, dropdowns and dialogs across every role.

2. The mill in one picture

Material moves through the mill in one direction. Each box below is a department, and each department has one role in the application. The arrow shows what is handed over.

Gate
Gateman registers the vehicle, photographs it and the bill. Two kinds: *unloading* (cotton arriving) and *loading* (yarn leaving).

→

Godown (raw material)
Godown Manager unloads, weighs each bale, prints a QR label and stores it by client (party).

→

Mixing
Bales are issued into a mixing lot. Own waste is added. The lot is completed.

Production
Supervisor records each shift on O/E, Card and Draw Frame sheets, and every DOFF (loose yarn off the O/E machine).

→

Packaging
Packaging Manager packs loose cones into bags for a client, and assigns bags to the loading vehicle.

→

Dispatch
The vehicle leaves the gate. Stock is reduced and the Admin sees it in the dashboard.

THE BALANCE THAT MANAGEMENT WATCHES

Raw material received = in godown + in mixing + produced (and packed) + dispatched + (waste and process loss). The Overview screen in the Admin console shows these figures side by side so any gap is visible immediately.

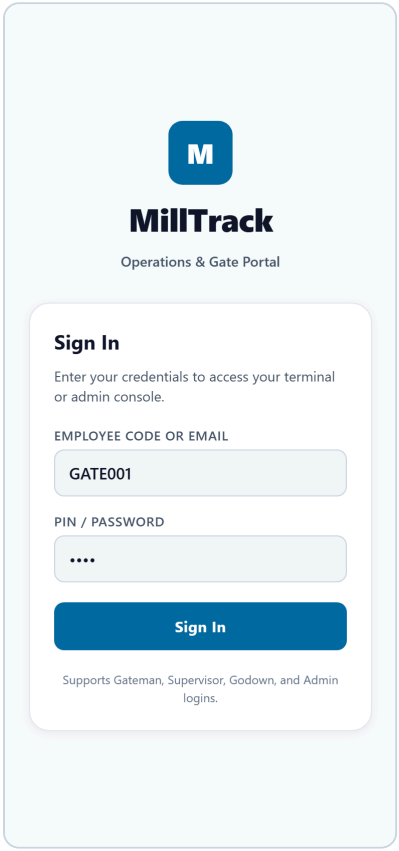
2.1 Who uses what

Role (login)	Typical person	What they do in the app	Home screen
Gateman GATEMAN	Security / weighbridge clerk	Registers vehicles entering and leaving, photographs vehicle and bill.	Vehicles list
RM Godown GODOWN_MANAGER_RM	Raw-material godown manager	Unloads vehicles, weighs bales, manages godown stock and mixing lots.	Unloading queue
Supervisor SUPERVISOR	Shift supervisor, production floor	Fills the shift production sheets and logs DOFFs.	Shift sheet
PKG Godown GODOWN_MANAGER_PKG	Packaging / finished-goods manager	Packs bags, tracks loose stock, loads vehicles, keeps the paper log.	Pack bag
Admin GM_MD_ADMIN	GM, MD or system administrator	Watches stock and production, manages users, pallis and the mixing recipe, opens permanent reports.	Admin console

3. Getting started (all roles)

3.1 Installing and signing in

1. Install the MillTrack app (APK file or Play Store build) on an Android phone.
2. Open the app and sign in with the **employee code** (or email) and **PIN / password** the Admin created for you.
3. The app takes you straight to the screen for your role. You never see other roles' screens.
4. If you are away from the app for a long time and your session expires, you are sent back to the login screen automatically. Nothing you saved is lost.



00-login

Login screen

Enter your **Employee Code (or email)** and your **PIN / password**, then sign in. If the details are wrong, or the phone cannot reach the server, a clear message appears above the button.

- The Admin creates every account. There is no self-registration.
- Roles decide which screens open: this is controlled by the server, not by the phone.
- The Gateman screen is additionally tied to a physical gate (see Section 4.4).

3.2 Common patterns you will see everywhere

- **Dropdowns** open as a sheet from the bottom of the screen with a search box. Pick an item to close it.
- **Confirmation dialogs** appear before any action that cannot be undone (finishing unloading, locking a shift, finalizing a day).
- **Messages**: a green or red banner appears at the top and disappears by itself. Inline red text appears next to the field that needs attention.
- **Sign out** is in the header of each role's screen.

4. Gateman

Gateman Registers every vehicle that comes in or goes out.

The gate is where traceability starts. If a vehicle is not registered here, nobody else in the mill can receive or load it. The Gateman records who came, with what, from which client, and takes two photographs as evidence.

Key result areas (KRA)

- Every vehicle entering and leaving is registered, correctly and promptly.
- Evidence is captured: vehicle photo (and bill photo where there is one).
- Vehicles are marked as exited when they leave, so the list shows only what is on the premises.
- Entry details are complete: vehicle number, party, driver, material, challan.

Suggested KPIs

- Vehicles registered per day (unloading and loading).
- Entries with a vehicle photo: target 100%.
- Entries with a bill photo, where a bill exists.
- Average time a vehicle stays on premises (entry to exit).
- Entries still open at end of day.
- Entries rejected or corrected because of wrong data.

ABOUT THE KRA / KPI LISTS

KRAs are what the role is accountable for. KPIs are measurements that can be taken from the data MillTrack already records. They are suggested starting points for the mill's own review meetings, not targets enforced by the software.

4.1 Daily flow

1. Open the app
The Vehicles list shows today's vehicles and their status.

→

2. New entry
Tap the plus button. Fill the form, take the photos, submit.

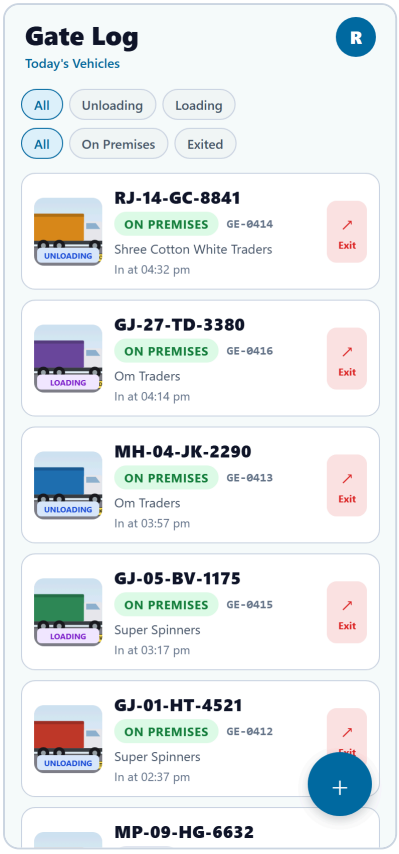
→

3. Vehicle appears
For unloading, it appears in the Godown Manager's queue at once. For loading, in the Packaging Manager's dispatch list.

→

4. Exit
When the vehicle leaves, tap **Exit** once (or **Mark as Exited** on its detail screen).

4.2 Screens

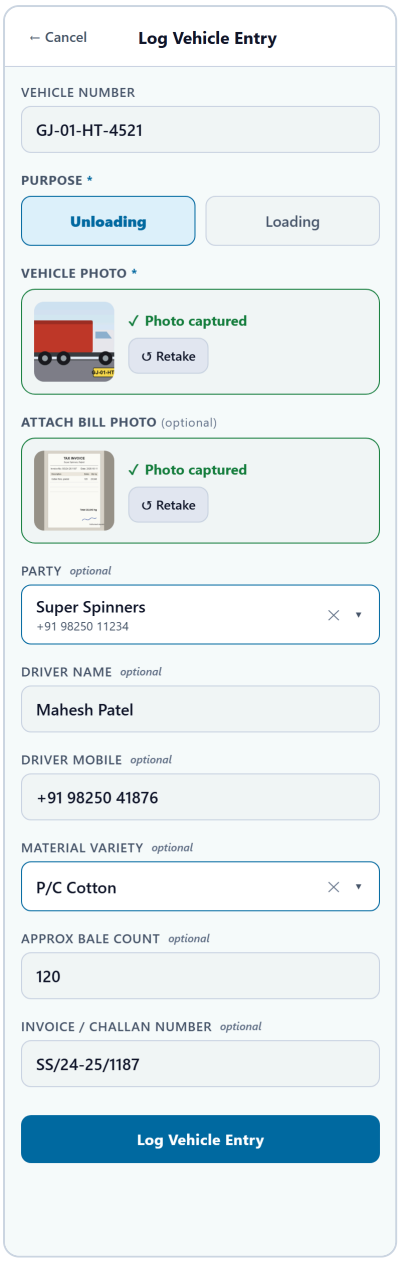


gate-01-vehicles

Vehicles (home)

Today's vehicles, newest first, with a status badge for each (on premises or exited). Two filter rows at the top narrow the list by purpose (unloading or loading) and by status (all, on premises, exited). Pull down to refresh.

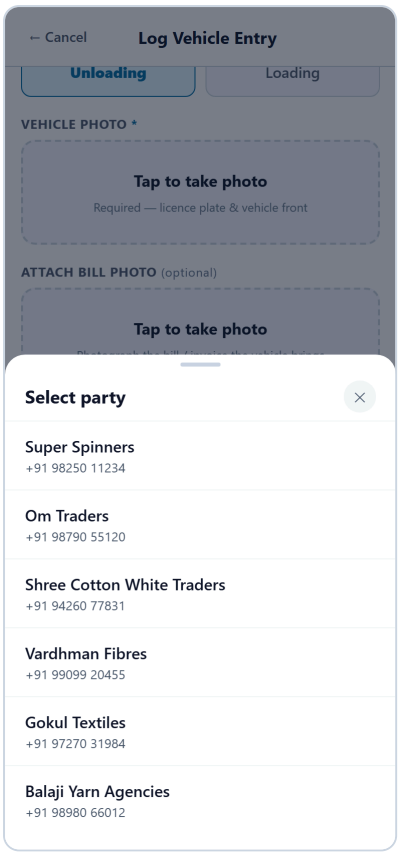
- **Plus button** starts a new entry.
- The red **Exit** button on a card marks a vehicle that is leaving, in one tap.
- Tap a card to open the full detail.
- Works with poor signal: entries made offline are queued and sent automatically when the connection returns.



gate-02-new-entry

New entry form

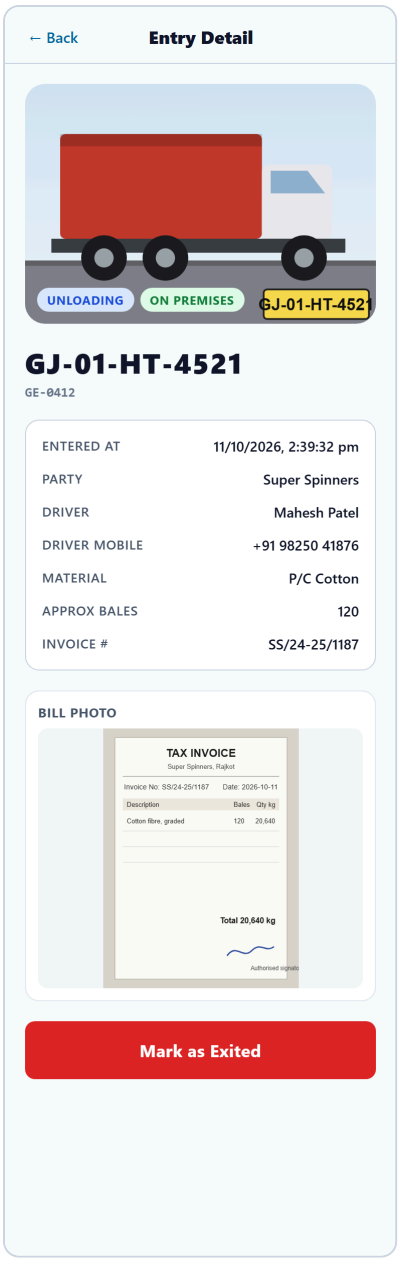
- **Purpose**: Unloading (cotton arriving) or Loading (yarn leaving).
- **Vehicle number**, **Party** (the client), **Driver name** and **mobile**.
- **Material variety** and **approximate bale count** (for unloading only).
- **Invoice / challan number**.
- **Vehicle photo** (required) and **bill photo** (optional). Photos are taken live with the camera; the gallery cannot be used, so a photo is always of the real vehicle.
- Tap **Submit**. A red message explains anything missing.



gate-03-party-dropdown

Choosing the party

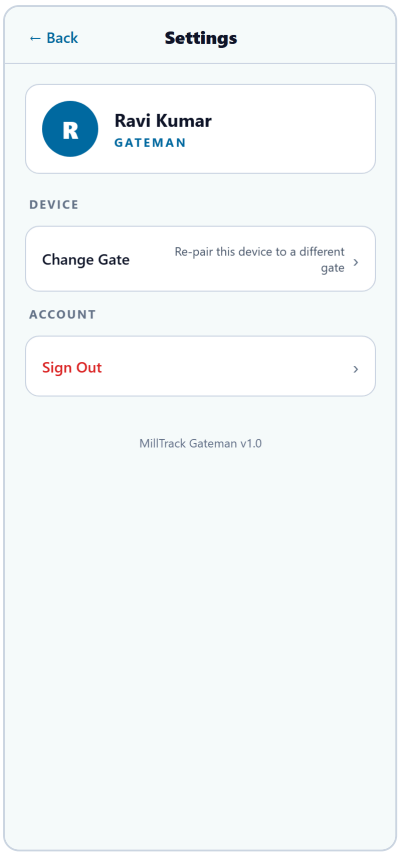
The party list opens as a bottom sheet with a search box, and stays above the keyboard so nothing is hidden. Parties are added by the Admin; the Gateman cannot invent new ones, which keeps client names consistent across the mill.



gate-04-vehicle-detail

Vehicle detail

The full record of one entry: number, party, driver, material, challan, the two photos, entry time and status. If the vehicle is still on the premises, a red **Mark as Exited** button is shown at the bottom.



gate-05-settings

Settings

Shows who is signed in and which gate this phone is paired with. **Change gate** re-pairs the phone (for example when a phone moves from the main gate to the weighbridge). **Sign out** ends the session.

TECHNICAL NOTES

- Photos are uploaded straight to cloud storage with a one-time **presigned URL** (`POST /gate/uploads/presign` for the vehicle photo, `/presign-bill` for the bill). The entry then stores only the resulting URLs.
- Entry: `POST /gate/entries` ; list: `GET /gate/entries` ; exit: `POST /gate/entries/{id}/exit` .
- If the network is down at submit time, the entry (with its photos) is stored in a local queue (`lib/offlineQueue.ts`) and retried when connectivity returns. Each entry has a client-generated id, so a retry can never create a duplicate.
- Camera: `expo-camera` , live capture only.

GOOD TO KNOW

If a vehicle is registered with the wrong purpose, the Godown or Packaging team will not see it in their list. Check the purpose before submitting.

5. RM Godown Manager (raw material)

RM Godown Receives cotton, controls the godown stock and builds the mixing lots.

This role owns the raw material from the moment a truck is unloaded until the cotton is mixed and handed to production. Every bale is weighed, labelled with a unique QR code, and recorded against the client (party) it belongs to.

Key result areas (KRA)

- Every unloaded bale is weighed and recorded accurately (gross, tare, net).
- The godown stock is correct and visible by party.
- Mixing lots follow the approved standard recipe.
- Lots are completed promptly so production can proceed.
- Mistakes are corrected openly (void a bale, update tare), never hidden.

Suggested KPIs

- Vehicles unloaded per day; average unloading time per vehicle.
- Bales weighed per day; total net kg received.
- Gap between the gate's approximate bale count and the bales actually weighed.
- Number of bales voided or tare corrections (lower is better).
- Godown stock by party vs. physical count.
- Mixing lots completed vs. started; share of lots matching the standard recipe.

ABOUT THE KRA / KPI LISTS

KRAs are what the role is accountable for. KPIs are measurements that can be taken from the data MillTrack already records. They are suggested starting points for the mill's own review meetings, not targets enforced by the software.

5.1 Daily flow

1. Unloading queue
Vehicles registered at the gate for unloading, waiting for you.

→

2. Set up the vehicle
Choose godown location, party, material, grade and tare. Done once per vehicle.

→

3. Weigh bales
Type each bale weight. The app saves it and prints a QR label.

→

4. Finish unloading
Confirm. The vehicle is closed and stock is updated.

5. Godown stock
See every bale by party. Add bales into a mixing lot.

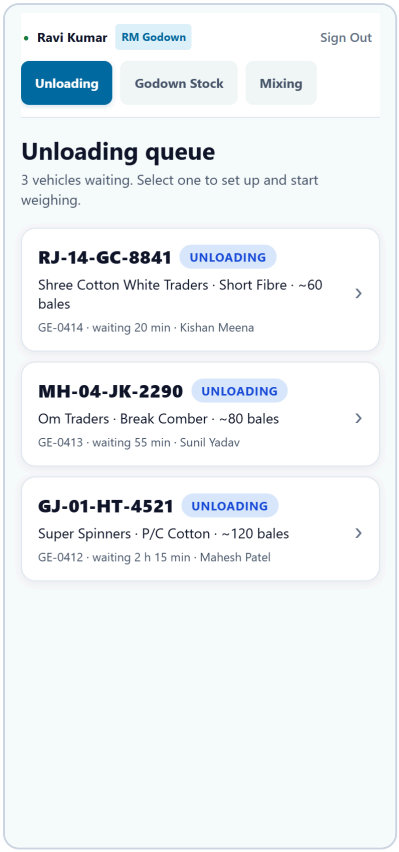
→

6. Mixing sheet
Bales appear in party boxes, own waste is typed in.

→

7. Mixing complete
Lock the lot. The system also files a permanent Mixing Sheet PDF for the Admin.

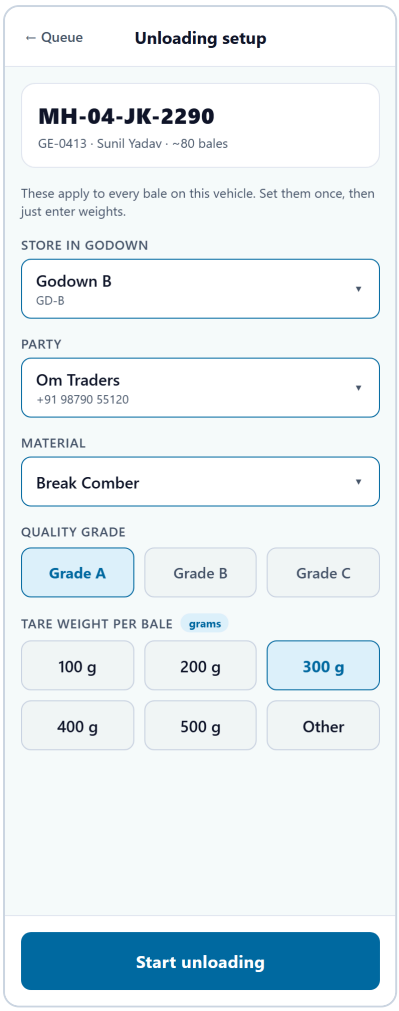
5.2 Unloading a vehicle



godown-01-queue

Unloading queue

Every vehicle the gate has registered for unloading that is waiting or in progress, with party, material, approximate bales and how long it has been waiting. Tap a vehicle to start. Pull down to refresh; the list also refreshes itself.

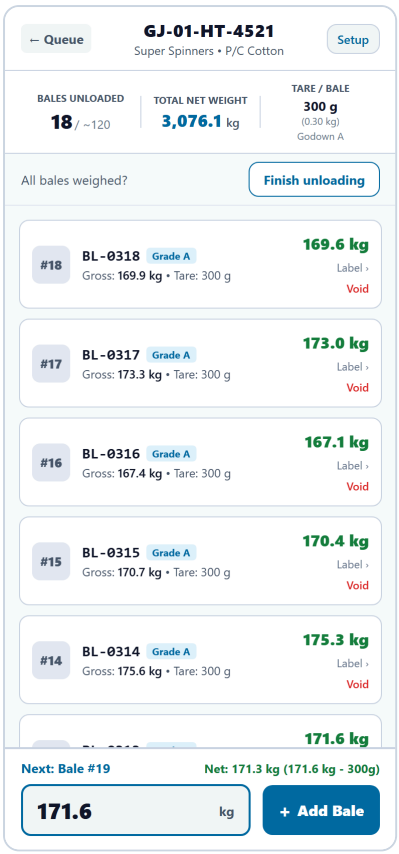


godown-02-unload-setup

Step 1: set the vehicle up

All the rules that apply to *every* bale on this vehicle are chosen once, so afterwards the worker only types weights.

- **Godown location** where the bales will be kept.
- **Party** and **material variety** (pre-filled from the gate entry).
- **Quality grade** A, B or C.
- **Tare weight per bale**: the weight of the wrapping and straps, in **grams**. Quick presets 100, 200, 300, 400, 500 g; choose **Other** to type a different value (anything above 500 g is allowed but flagged).
- The app remembers your last choices so the next vehicle starts pre-filled. Reopening a vehicle that already has bales skips straight to weighing.



godown-03-weighing

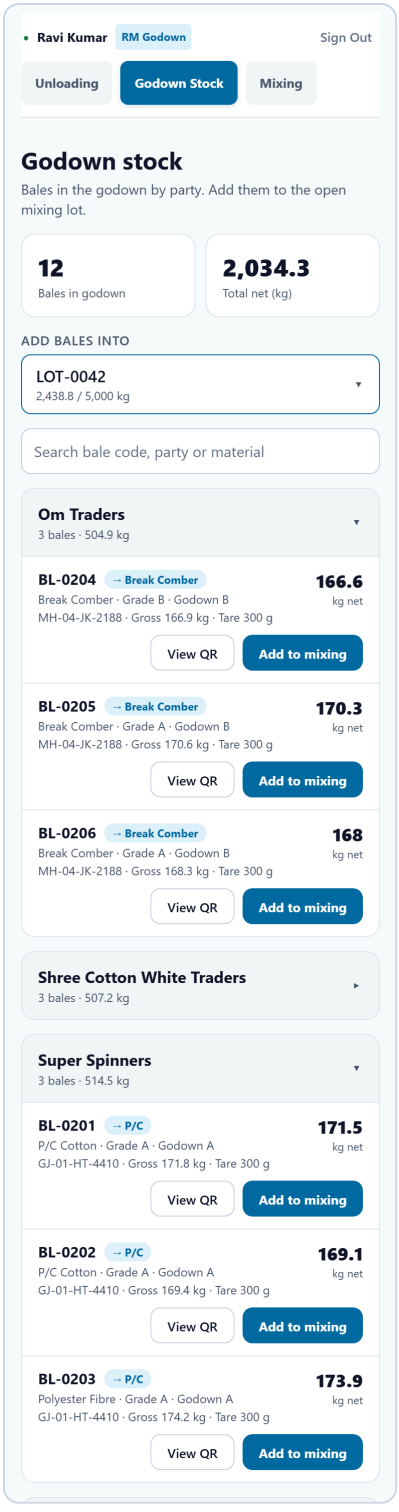
Step 2: weigh the bales

- Type the bale's **gross weight in kg** and save. The app shows the **net weight** (gross minus tare) and creates a unique **bale code / QR**.
- The list of bales for this vehicle, with running totals, stays on the screen.
- **Void** a bale entered by mistake (it stays in the history, marked void).
- **Update tare** changes the tare for the bales already weighed, after a confirmation, if the wrong preset was used.
- **Finish unloading** closes the vehicle after a confirmation. After this no more bales can be added.

TECHNICAL NOTES

- Bale: `POST /godown/bales` (client-generated id, so a retry is safe); list for a vehicle: `GET /godown/entries/{id}/bales` ; tare update: `PATCH /godown/entries/{id}/bales/tare` ; void: `POST /godown/bales/{id}/void` ; finish: `POST /godown/entries/{id}/finish-unloading` .
- Weights are decimal strings (never floating point) to avoid rounding drift. Tare is entered in grams in the UI and stored in kg.
- Printing: `POST /godown/bales/{id}/print` asks the server to print the label. The on-device label hook (`lib/printLabel.ts`) is currently a placeholder awaiting the printer decision.

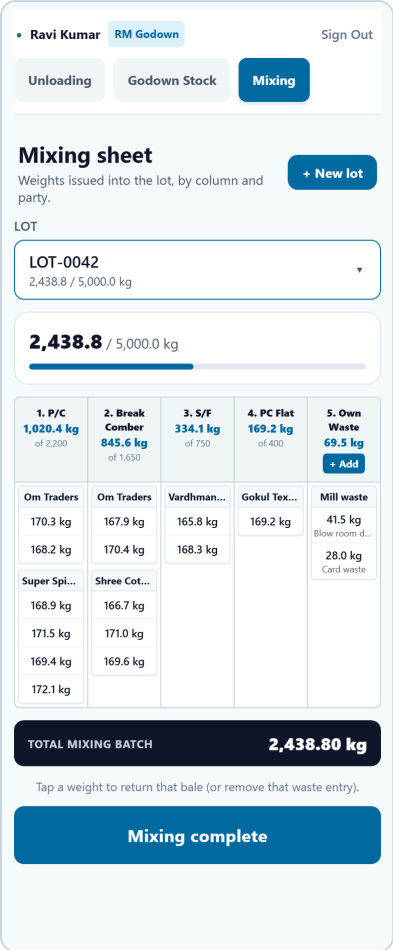
5.3 Godown stock and mixing



godown-04-stock

Godown stock (party-wise)

Every bale currently in the godown, grouped by party, with bale count and net kg per party. Choose the open mixing lot in **Add bales into** at the top, then tap **Add to mixing** on a bale and pick which mixing column it goes into. The party stays attached to the bale, so the mixing sheet always shows where each kilogram came from.

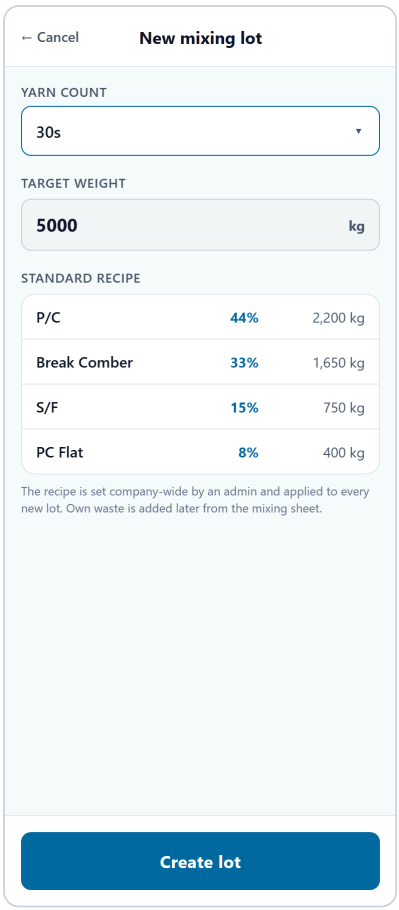


godown-05-mixing

Mixing sheet

The blow-room master view of one lot. Columns come from the Admin-set mixing categories. **Fibre columns** show the scanned bales, grouped in party boxes with net kg. **Waste columns** hold the mill's own waste, typed in with an optional note. All columns share the full screen width, so there is no sideways scrolling.

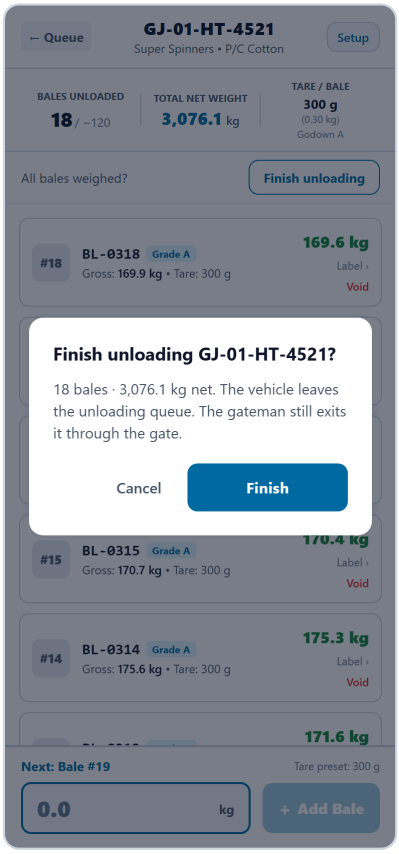
- Tap a weight to return that bale to stock, or remove a waste entry.
- Items can only be added or removed while the lot is **Building**.
- **Mixing complete** locks the lot. It is the only lot action this role has.



godown-06-new-lot

New mixing lot

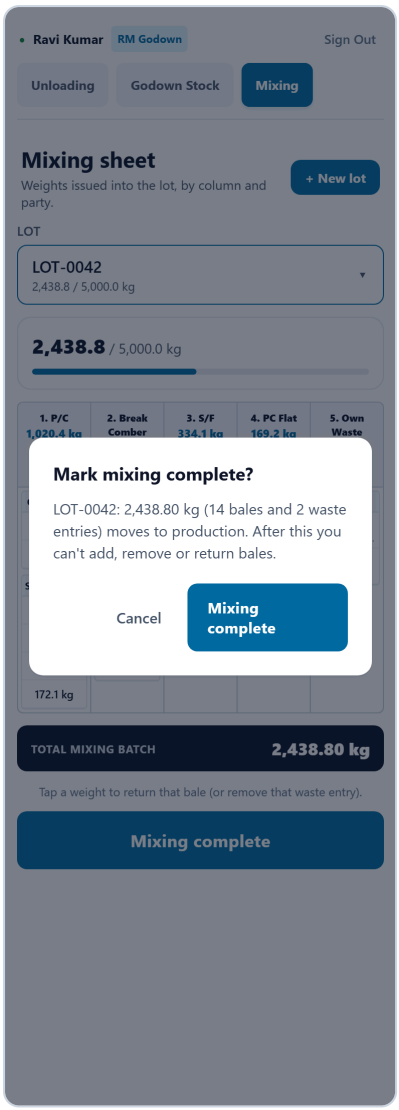
Pick the **yarn count** and the **target weight**. The recipe (the percentage of each column) is the company-wide standard set by the Admin; the server applies it and this screen only previews it, so lots cannot drift away from the approved mix.



godown-07-finish

Confirmations

Finishing unloading and completing a mixing lot both ask for confirmation, because they close the work. The dialog names the vehicle or lot so the wrong one is not closed by accident.



godown-07b-mixing-complete

Mixing complete

Completing a lot asks for confirmation and names the lot. After it, the lot is locked and the Mixing Sheet PDF is filed.

TECHNICAL NOTES

- Stock: `GET /godown/stock-bales` ; lots: `GET/POST /godown/mixing-lots` , `GET /godown/mixing-lots/{id}` .
- Add/return a bale: `POST /godown/mixing-lots/{id}/scan` and `DELETE .../scan/{scanId}` . Waste: `POST .../waste` , `DELETE .../waste/{wasteId}` .
- Complete: `POST /godown/mixing-lots/{id}/complete` . The backend then generates the Mixing Sheet PDF automatically; the app does nothing extra (see Section 9).
- Scans and waste are accepted only while the lot is BUILDING; the server enforces this.

6. Supervisor

Supervisor Records production for each shift and every DOFF.

The Supervisor's screen is the digital version of the paper "Production Sheet Day". There is one sheet per process (O/E, Card and Draw Frame) and one row per machine. At the end of the shift the supervisor locks the shift, which freezes it.

Key result areas (KRA)

- Every machine's output, efficiency and stoppages are recorded for every shift.
- Every DOFF is logged at the time it is taken.
- The shift is locked on time.
- Errors found later are corrected visibly, with a reason.

Suggested KPIs

- Shifts locked on time (and at all) vs. shifts worked.
- Machine rows filled vs. active machines.
- Production kg per shift and per process; efficiency %.
- Total breakage, run time, idle time and JSI % trends.
- DOFFs per machine per shift; DOFF interval vs. the expected ~20 hours.
- Corrections made after lock (lower is better).

ABOUT THE KRA / KPI LISTS

KRAs are what the role is accountable for. KPIs are measurements that can be taken from the data MillTrack already records. They are suggested starting points for the mill's own review meetings, not targets enforced by the software.

6.1 Daily flow

1. Open the app
The app asks the server for the current shift (A starts 7 am, B starts 7 pm). The phone's own clock is never used.

→

2. Fill the sheets
O/E, Card, Draw Frame: one row per machine. Save.

→

3. Log DOFFs
Whenever a DOFF is taken, log it on the DOFF tab. Loose yarn enters the loose stock.

→

4. Lock the shift
Confirm. The shift is frozen and a Production Sheet PDF is filed for the Admin.

6.2 Screens



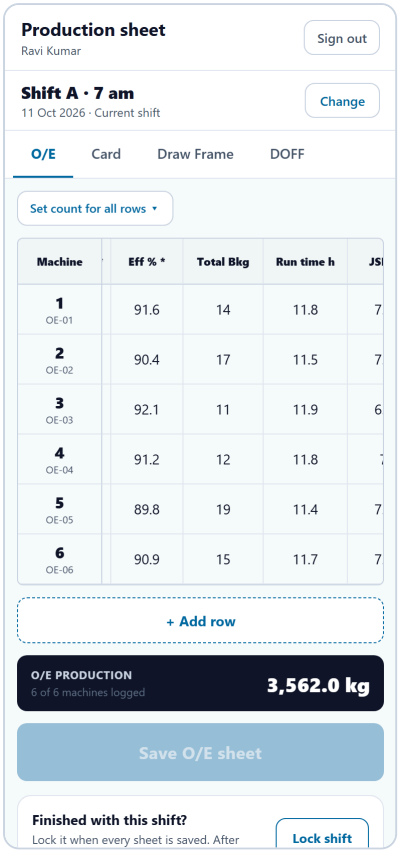
sup-01-oe

O/E sheet

The shift bar at the top shows the shift in use and its date. Under it, the three process tabs (O/E, Card, Draw Frame) and DOFF. Each machine is a row and the columns mirror the paper sheet.

Column	Meaning
Count	Yarn count being made
D.E.L	Delivery / machine reading
Rotor speed	Rotor RPM (O/E)
Production kg	Output this shift (required)
Eff %	Efficiency, 0 to 100 (required)
Total Bkg	Total breakages
Run / Idle time h	Hours running / stopped (max 12)
JSI %	Quality index
Detention	Free-text note on stoppages

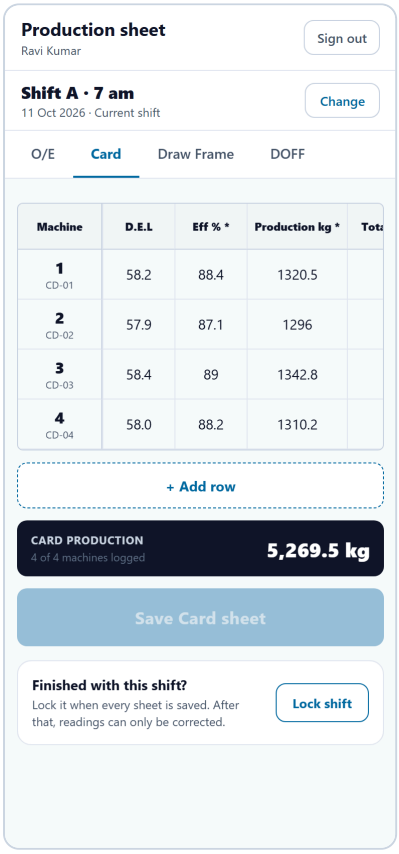
Changed rows turn yellow until saved; a red message marks a row with a problem.



sup-01b-oe-right

O/E sheet, scrolled to the right

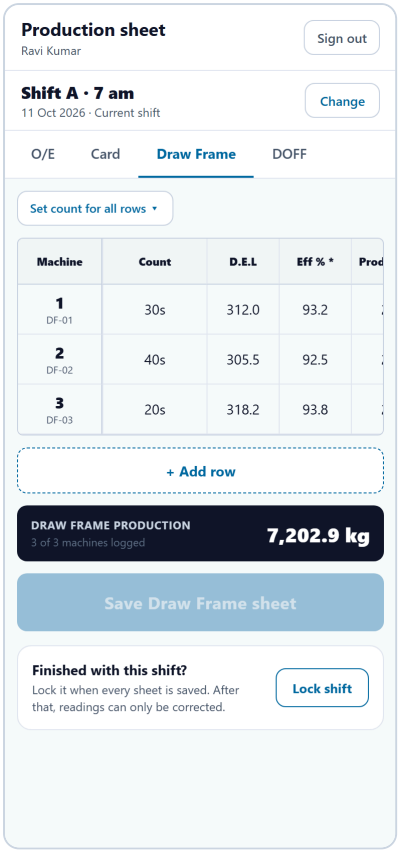
The sheet is wider than a phone screen, like the paper register, so it scrolls sideways. This view shows the right-hand columns (Production kg, Eff %, breakage, run time, JSI %, detention). The machine number stays visible on the left.



sup-02-card

Card sheet

Same layout with the columns the Card section uses (D.E.L, Eff %, Production kg, Total Bkg, Run time, Idle time, Detention).



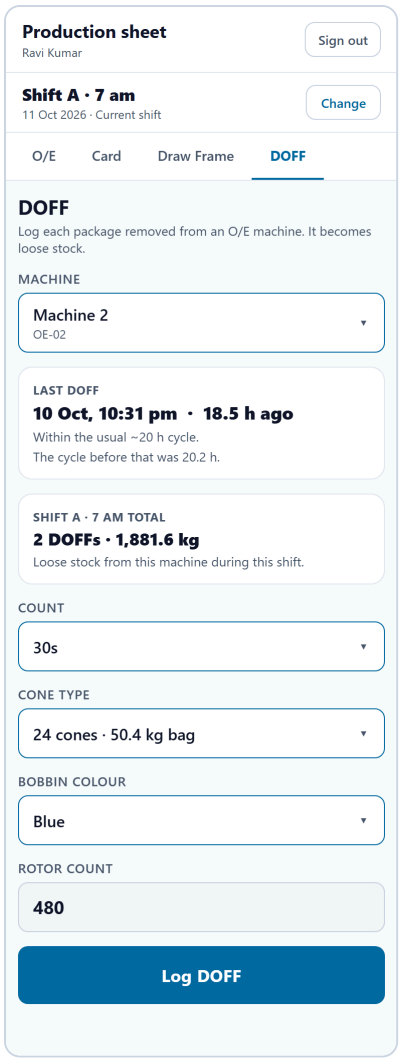
sup-03-draw

Draw Frame sheet

Count, D.E.L, Eff %, Production kg, Total Bkg, Run time and Detention for each Draw Frame.

MACHINES ARE ADDED WHERE THEY ARE USED

Machine numbers are not fixed in advance. If a machine is missing, the supervisor adds it directly on the sheet (**Add machine**). A machine added by mistake can be deleted with **Delete**, but only while nothing has been recorded for it; otherwise the server refuses and the message says so.

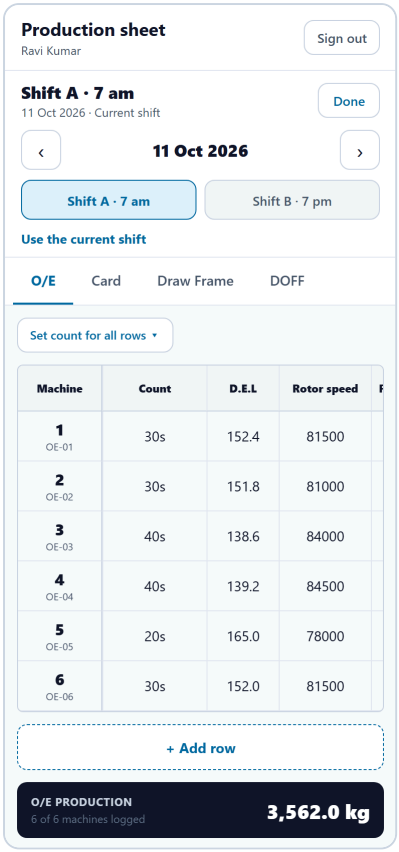


sup-04-doff

DOFF tab

A DOFF is the moment a full package of loose yarn is taken off an O/E machine. Choose the **O/E machine**, the **count**, **cone type** and **bobbin colour**, confirm the **rotor count** (pre-filled for the machine) and tap **Log DOFF**; the server works out the loose-stock weight. The card shows the **total for this shift**: the number of DOFFs and the loose kg from that machine. The screen also shows time since the machine's last DOFF and warns when it is far from the expected ~20 hours (below 16 or above 24), which usually means a missed or duplicate entry.

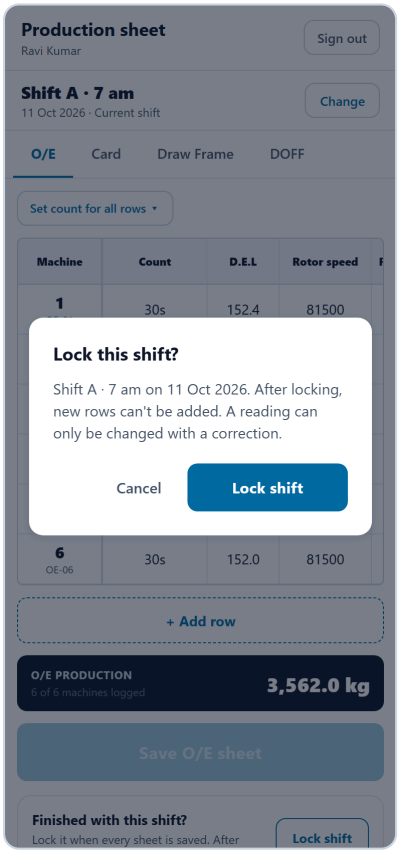
This loose yarn is what the Packaging Manager later sees in the loose-stock pool.



sup-06-change-shift

Changing the shift (exception)

Normally the app uses the current shift from the server. **Change** lets the supervisor open another shift to catch up on an earlier one: choose the date and Shift A or B. A banner offers **Switch** if the shift changes while the app is open, so unsaved work is never lost silently.



sup-05-lock

Lock shift

Lock shift asks for confirmation, then freezes the shift. After locking, saved rows cannot be overwritten; they can only be **corrected**: a correction creates a new entry that points to the original and must carry a short note explaining why. The Admin's permanent Production Sheet PDF is generated by the backend at lock time.

TECHNICAL NOTES

- Current shift: `GET /supervisor/current-shift` (server owns the 7 am / 7 pm boundary and time zone). Machines: `GET/POST /supervisor/machines`, `DELETE /supervisor/machines/{id}` (409 once used).
- Rows: `POST /supervisor/shift-logs/{process}`; lock: `POST /supervisor/shift-submissions`; DOFF: `POST /supervisor/doff`, last: `GET /supervisor/doff/last/{machine}`, shift total: `GET /supervisor/doff/total/{machine}`.
- Rows carry client-generated ids; a correction sends `correction_of_id` and `correction_note`.
- Run/idle time are entered in decimal hours. If the mill writes them as H.MM (for example 7.30 for 7 h 30 m) the entry rules should be changed; this is an open item for confirmation.

7. PKG Godown Manager (packaging and dispatch)

PKG Godown Packs yarn cones into bags and loads vehicles.

The loose cones that production produces become sellable stock only when they are packed into bags. The Packaging Manager weighs and labels each bag for a client, assigns bags to vehicles that arrive to load, and keeps the paper log of the day.

Key result areas (KRA)

- Every bag is packed against the right client, count, colour and cone type, with its true weight.
- Vehicles are loaded with exactly the bags promised.
- Loose stock is visible and does not pile up.
- The day's paper log is entered and finalized.

Suggested KPIs

- Bags packed per day; kg packed.
- Loose stock (cones/kg) at end of day.
- Dispatches completed; bags dispatched vs. planned per vehicle.
- Average loading time per vehicle.
- Paper log completed and finalized every day.
- Closing stock check: closing = loose + packed - out (soft hint shown by the app).

ABOUT THE KRA / KPI LISTS

KRAs are what the role is accountable for. KPIs are measurements that can be taken from the data MillTrack already records. They are suggested starting points for the mill's own review meetings, not targets enforced by the software.

7.1 Daily flow

1. Pack bags
Choose count, client + colour, bobbin colour, cone type; weigh; save. A unique bag code is returned.

→

2. Dispatch
A vehicle arrives to load. Open a dispatch, declare what should go, assign the packed bags.

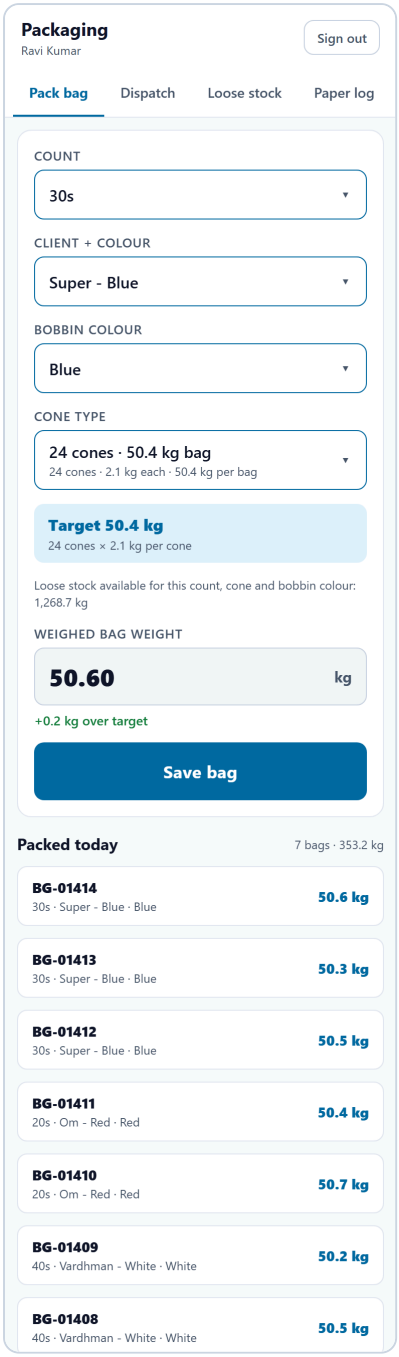
→

3. Paper log
Type the paper register's numbers for the day in the sheet.

→

4. Finalize
At end of day, finalize. A permanent Packaging Sheet PDF is filed for the Admin.

7.2 Screens



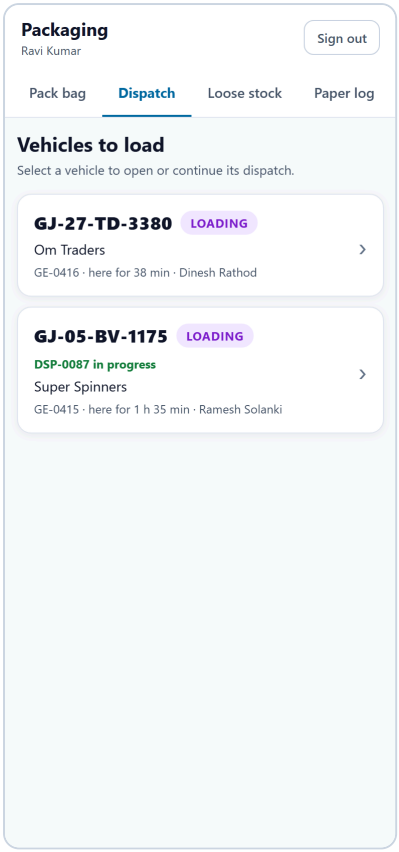
pkg-01-pack

Pack bag

One bag per entry. Choose **Count**, **Client + colour** (the "Palli"), **Bobbin colour** and **Cone type**, enter the weight and tap **Save bag**. The server returns the bag's unique code. Your selections are kept, so the next bag of the same kind needs only a new weight.

HOW WEIGHTS WORK

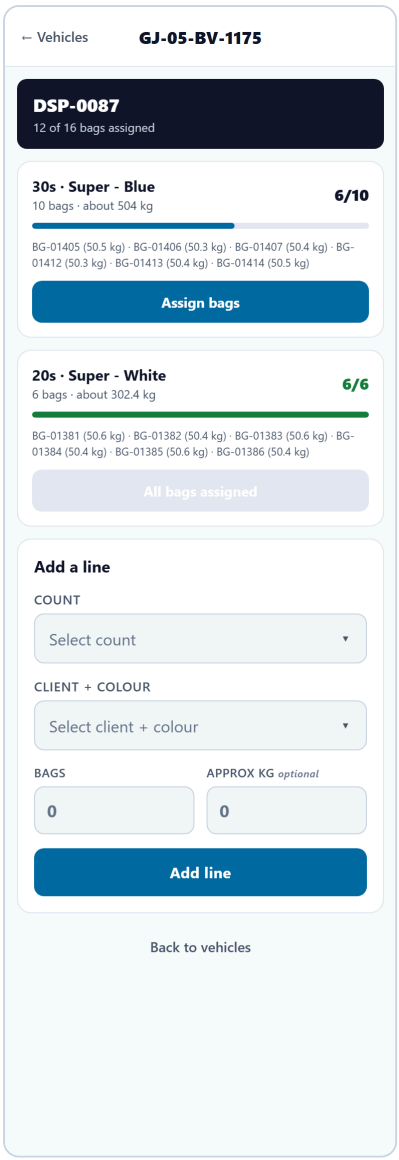
A cone type is "N cones per bag at X kg per cone". The target bag weight is N times X. The screen shows this target and how far the weighed bag is over or under it, so a wrongly weighed bag is noticed before saving.



pkg-02-dispatch

Dispatch: vehicles to load

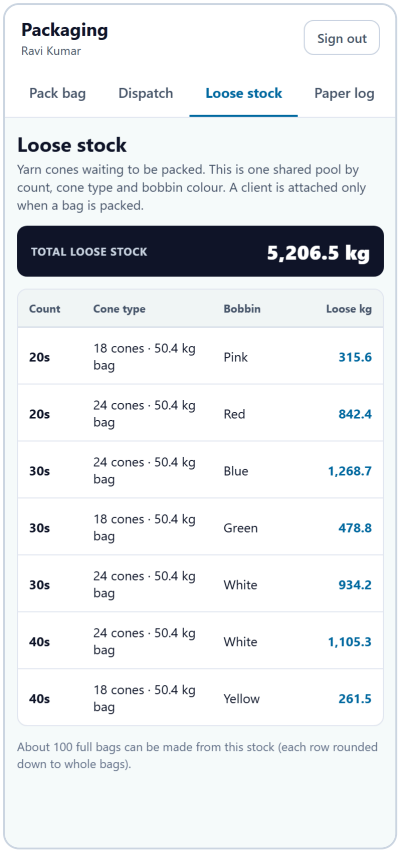
Vehicles registered at the gate for **loading** that are still on the premises. Tap one to open its dispatch.



pkg-03-dispatch-detail

Dispatch detail

Declare what should go on the vehicle as lines (count + client/colour + number of bags), then assign specific packed bags to each line. Everything shown comes from the server, so reopening the screen later, or on another phone, continues exactly where it was left.



pkg-04-loose

Loose stock

The shared pool of yarn cones that production has made and that are not yet in a bag. Nothing here belongs to a client yet: a client is attached only when a bag is packed.



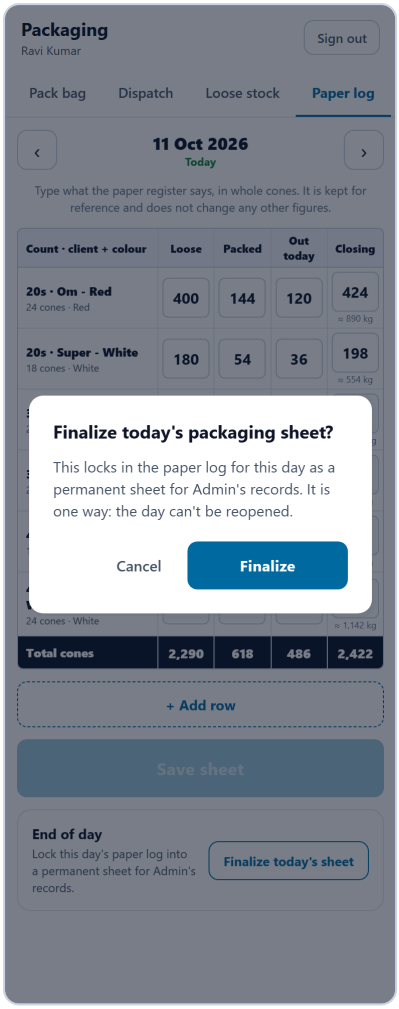
pkg-05-paper-log

Paper log

The paper register as a sheet, one grid per day. One row per count + client/colour + cone type + bobbin colour. Type the four numbers directly into the boxes, in whole cones:

- **Loose**: loose cones.
- **Packed**: cones packed.
- **Out today**: cones dispatched.
- **Closing**: cones left at the end.

Use **+ Add row** to add a combination; its first box takes the cursor at once. Changed rows turn yellow; **Save sheet** saves them. A soft hint (never blocking) mentions when closing is not loose + packed - out, or packed is not whole bags. The paper log is kept for reference and does not change any other stock figure.



pkg-06-finalize

Finalize the day

At the end of the day, **Finalize today's sheet** (or **Finalize this day** when another date is chosen for a late entry) freezes that day's paper log into a permanent Packaging Sheet PDF for the Admin. It asks for confirmation first. The button is disabled while there are unsaved changes, so the PDF cannot miss edits. Pressing it again for the same day is harmless.

TECHNICAL NOTES

- Pack: `POST /packaging/records` ; available bags: `GET /packaging/records/available` ; loose stock: `GET /packaging/loose-stock` .
- Dispatch: `GET /packaging/loading-vehicles` , `POST/GET /packaging/dispatch-entries` , `POST .../{id}/lines` , `.../dispatch-lines/{id}/bags` .
- Paper log: `PUT/GET /packaging/paper-log` (saving a row again overwrites it; rows cannot be deleted). Finalize: `POST /packaging/finalize-day` with `{"finalized_date":"YYYY-MM-DD"}` , response 201.
- Palli = client + bobbin colour, created by the Admin and delivered with the lookup lists.

8. Admin / GM / MD

Admin Sees the whole mill, manages the master data and owns the permanent records.

The Admin console is a set of tabs along the top. It answers the questions management asks every day: how much came in, where is it, how much did we make, how much left, and who did what.

Key result areas (KRA)

- Material is conserved: what came in is accounted for at every stage.
- The right people have access; leavers are deactivated.
- Master data (clients, pallis, standard recipe) is correct.
- Permanent records exist for every shift, lot and packaging day.

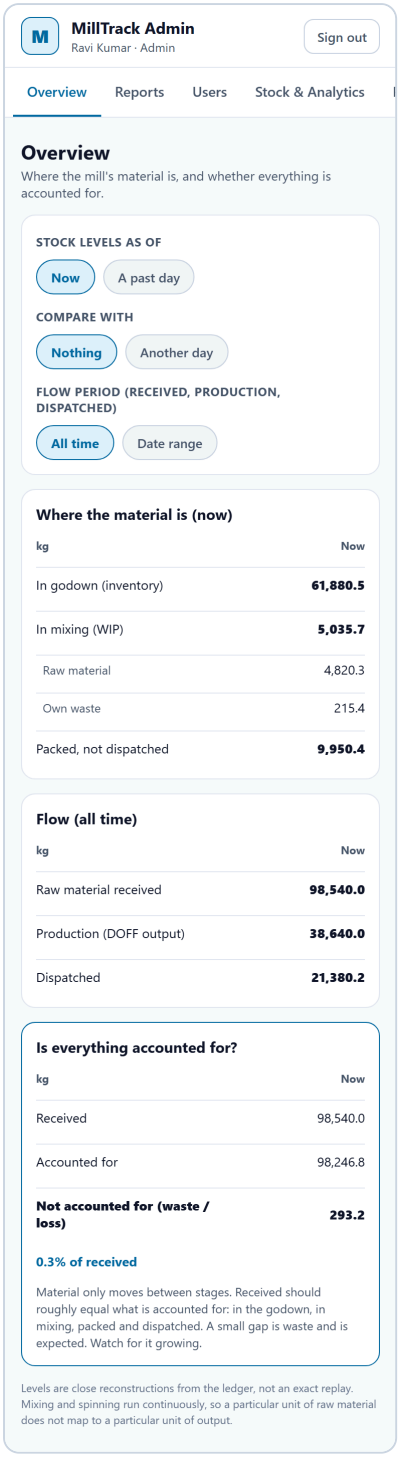
Suggested KPIs

- Raw material received, in godown, in mixing, packed, dispatched (kg) and the unexplained gap.
- Production kg per period.
- Reports generated vs. expected (one per shift, lot and day).
- Active users by role; accounts without activity.
- Time to resolve a trace request.

ABOUT THE KRA / KPI LISTS

KRAs are what the role is accountable for. KPIs are measurements that can be taken from the data MillTrack already records. They are suggested starting points for the mill's own review meetings, not targets enforced by the software.

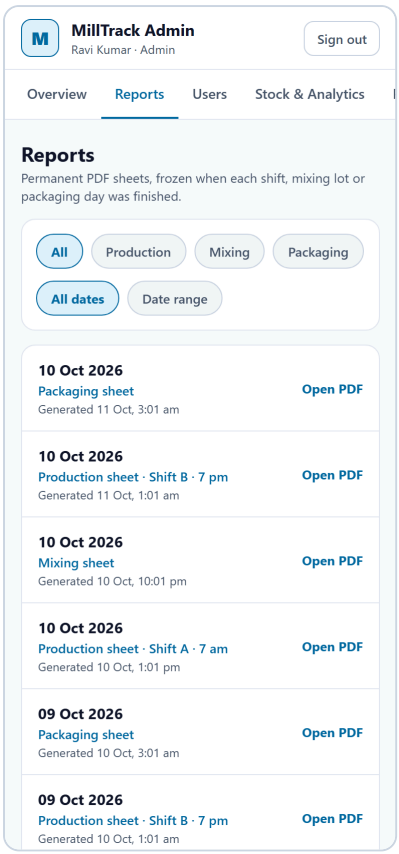
8.1 Screens (tabs)



adm-01-overview

Overview: the conservation dashboard

Tiles for **raw material received**, **in godown**, **in mixing** (split into raw material and own waste), **packed but not dispatched**, **production (DOFF output)** and **dispatched**. Choose **Now** or **A past day**; add a comparison day to see differences; and choose **All time** or a **Date range** for the flow figures. At the bottom, **Is everything accounted for?** compares what was received with what is accounted for (godown + mixing + packed + dispatched) and shows the remainder as waste or process loss, with its percentage of the received material.

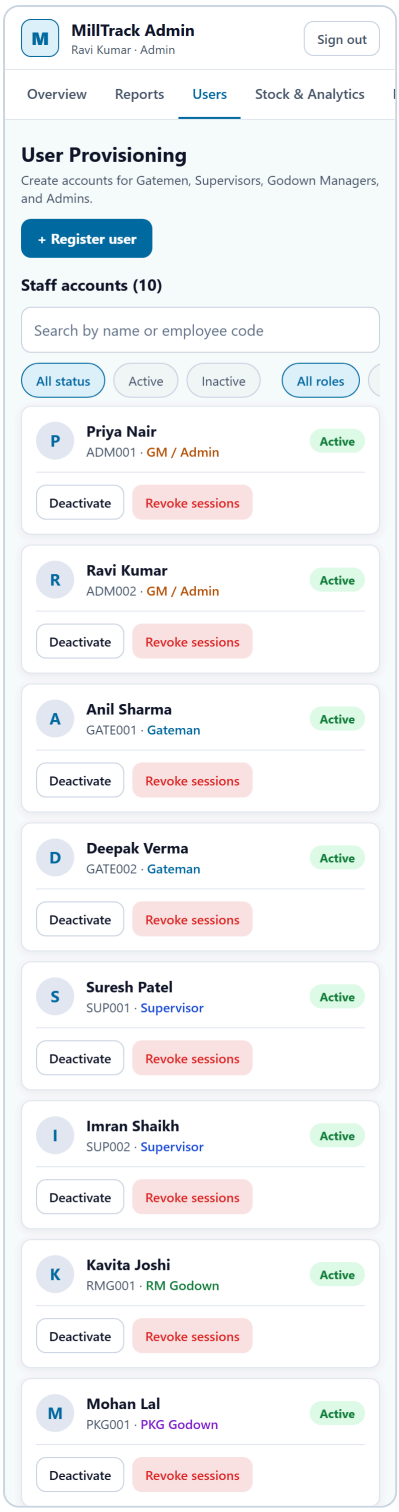


adm-02-reports

Reports: permanent PDF sheets

Every generated sheet, newest first: **Production** (one per locked shift), **Mixing** (one per completed lot) and **Packaging** (one per finalized day). Filter by type and date range. Each row shows the date, type (and shift for Production) and when it was generated. Tap a row to open the PDF.

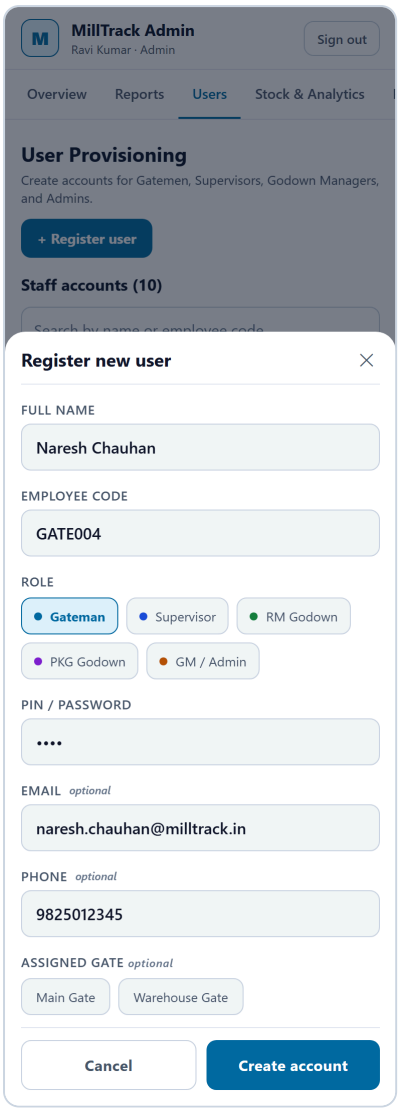
- The sheets are frozen at the moment of generation. Even if a figure is later corrected in the system, the PDF shows what was true then.
- There is deliberately **no edit and no delete** anywhere on this screen.
- File links are time-limited, so the list is fetched fresh every time the tab opens, and refreshed before opening if it is older than 10 minutes.



adm-03-users

Users

All accounts with their role and status. Filter by role. For each user: **Activate / Deactivate**, and **Revoke sessions** to sign the person out of every phone at once (for example when a phone is lost).

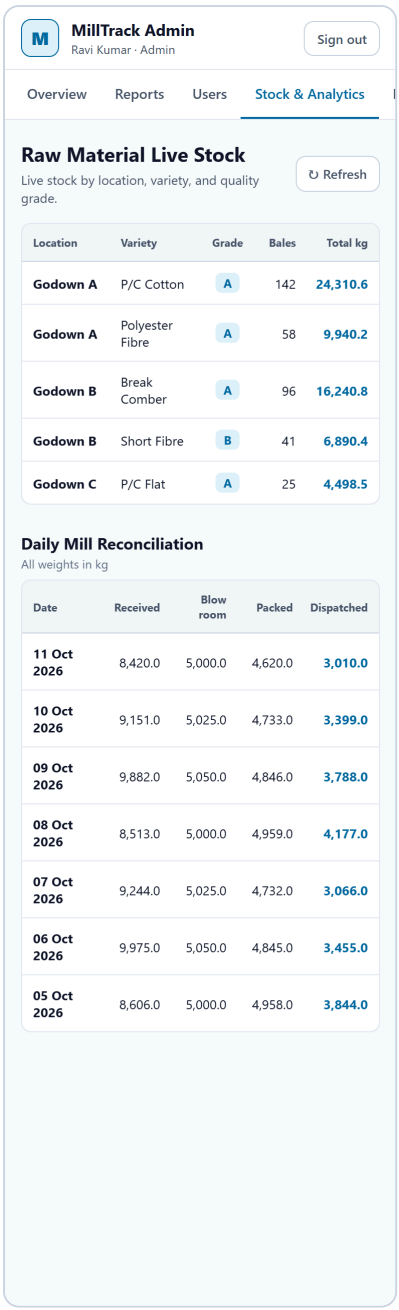


adm-04-register-user

Register a user

Enter the full name, employee code, PIN / password (at least 4 characters), email, phone and choose the role, optionally the assigned gate (for Gatemen), then tap **Create account**. A green message confirms the user was added; a red message explains problems such as "already registered".

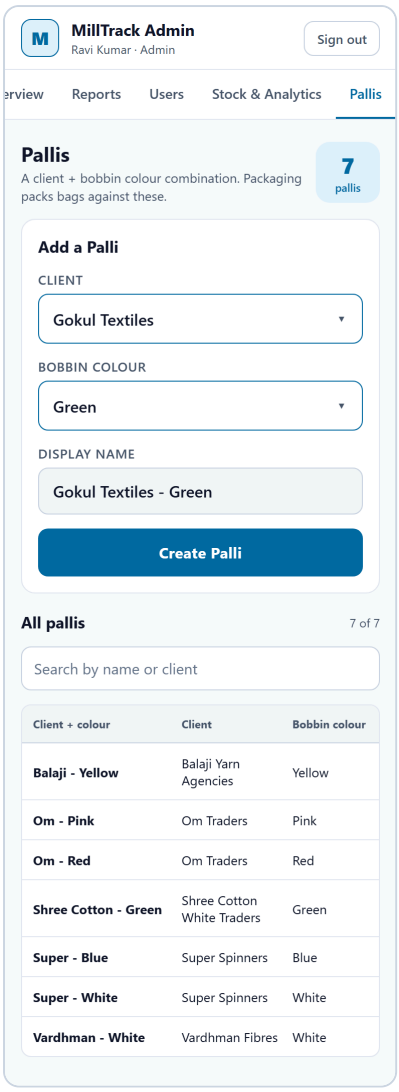
Role	Gives access to
Gateman	Gate entry, photos, exit
Supervisor	Shift sheets, DOFF
RM Godown	Unloading, stock, mixing
PKG Godown	Packing, dispatch, paper log
GM / Admin	Admin console



adm-05-analytics

Stock & Analytics

Two tables with aligned columns: **Raw Material Live Stock** (by godown location, variety and grade: bales and total kg) and **Daily Mill Reconciliation** (one row per day: received, in mixing, packed, dispatched).



adm-06-pallis

Pallis

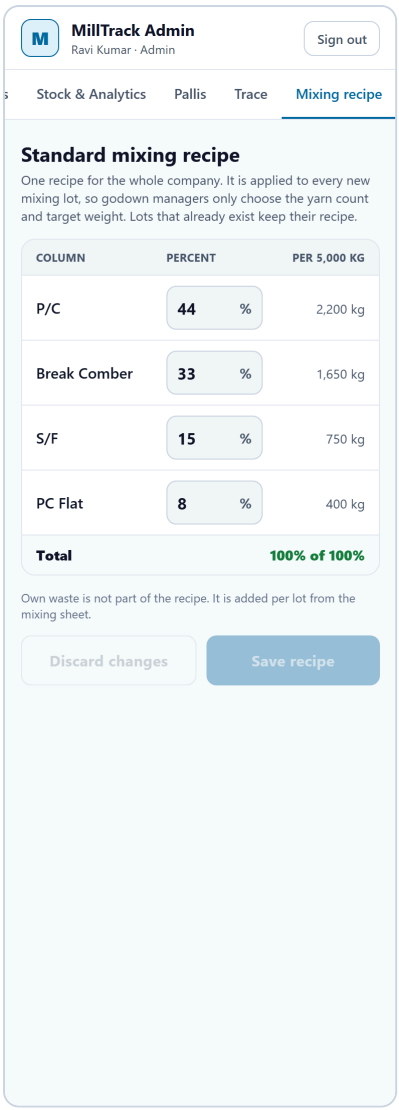
A Palli is a client plus a bobbin colour; packaging packs bags against it. Add a new one by choosing the client and colour (the display name is suggested, e.g. "Super - Blue"). The list shows every Palli with its client and colour, and the total count.



adm-07-trace

Trace

Enter any code (a bale QR code, a DOFF id and so on) to see its chain-of-custody record: the bale, the gate entry it arrived on, the mixing lot it went into, the bags made downstream and the dispatches. It is shown as a structured record (for example a bale code such as BL-0301).



adm-08-recipe

Mixing recipe

The standard mixing recipe for the whole mill: the percentage of each mixing category. Godown Managers cannot change it; every new lot gets this recipe from the server. Save with **Save recipe**; a confirmation dialog appears first.

TECHNICAL NOTES

- Stock summary: `GET /admin/analytics/stock-summary` (with `as_of_date` or `start_date` / `end_date`). Users: `GET/POST /admin/users`, `PATCH /admin/users/{id}`, `POST /admin/users/{id}/revoke`.
- Pallis: `POST /admin/pallis`. Recipe: `PUT /admin/mixing-recipe` (stored in the lookups). Trace: `GET /admin/trace/{code}`.
- Reports: `GET /admin/reports?report_type=&start_date=&end_date=`, newest first.
- The Pallis list comes from the active pallis in the lookups because there is no separate admin list endpoint.

9. Permanent PDF sheets (how reports are made)

Management wanted something close to the paper registers: a document for each piece of finished work that never changes. The backend now produces a real PDF for each of three events and stores it.

Sheet	Created when	Who triggers it	What the app does
Production Sheet	A supervisor locks a shift	Supervisor, **Lock shift**	Nothing new. It already existed.
Mixing Sheet	A mixing lot is completed	RM Godown, **Mixing complete**	Nothing new. It already existed.
Packaging Sheet	A packaging day is finalized	PKG Godown, **Finalize today's sheet**	New button on the Paper log screen.

Work is finished
A shift is locked, a lot is completed, or a day is finalized.

→

Backend makes the PDF
Generated and stored automatically. If it ever fails, the action itself still succeeds.

→

Admin opens it
Reports tab: filter, tap, read the PDF. Read-only.

10. A day in the life: the complete flow

This follows one lot of cotton and one order of yarn through the mill.

Step	Who	Screen	What is recorded	What happens next
1	Gateman	New entry	Vehicle KA-01-AB-1234 arrives with cotton for Super Spinners; vehicle + bill photos.	The vehicle appears in the Godown unloading queue.
2	RM Godown	Unloading setup, Weighing	Godown, party, material, grade, tare (e.g. 200 g). 24 bales weighed, QR labels printed.	Bale stock increases under Super Spinners.
3	RM Godown	Finish unloading	Vehicle closed.	The Gateman can mark it as exited.
4	RM Godown	New lot, Godown stock, Mixing sheet	A 20s lot at the target weight; bales issued from stock, own waste typed in.	Mixing complete locks the lot; a Mixing Sheet PDF is filed.
5	Supervisor	Shift sheets	O/E, Card and Draw Frame readings for the shift.	Lock shift freezes it; a Production Sheet PDF is filed.
6	Supervisor	DOFF	A DOFF from an O/E machine.	Loose yarn enters the loose-stock pool.
7	PKG Godown	Pack bag	Loose cones packed into a bag for a client; bag code returned.	Loose stock goes down, packed stock goes up.
8	Gateman	New entry (Loading)	A vehicle arrives to load.	It appears in the Dispatch list.
9	PKG Godown	Dispatch detail	Lines declared; packed bags assigned to the vehicle.	Packed stock goes down, dispatched goes up.
10	PKG Godown	Paper log, Finalize	The day's paper register typed and finalized.	A Packaging Sheet PDF is filed.
11	Admin	Overview, Reports	Reviews the dashboard and opens the sheets.	Any gap in the balance is visible straight away.

11. Rules of thumb for every role

- Enter data when the work happens, not at the end of the shift.
- If a message in red appears, read it: it says exactly what is wrong.
- Do not share your login. Every record carries the name of the person who entered it.
- If the phone is lost, tell the Admin at once so the sessions can be revoked.
- Mistakes are fixed through the app's correction features (void a bale, correct a locked row). Do not write the same number twice to "balance" it.

12. Technical overview

Area	Choice
Client app	Expo SDK 57, React Native, TypeScript, React Compiler enabled
Navigation	Expo Router (file-based). One route group per role: `(gate)`, `(godown)`, `(supervisor)`, `(packaging)`, `(admin)`
Data layer	TanStack Query v5 for caching, refetching and mutations; AppState is wired into the focus manager so data refreshes when the app returns to the foreground
Backend	FastAPI service. The contract is in `ApiDocumentation.json`. Base URL from `EXPO_PUBLIC_API_BASE_URL`
Auth	Bearer token from `POST /auth/login`, stored in `expo-secure-store`; a 401 anywhere returns the user to Login
Storage	SecureStore (token, role), AsyncStorage (last choices, offline queue)
Camera and QR	`expo-camera` (live capture only), `react-native-qrcode-svg`
IDs	`react-native-uuid`: client-generated UUID v4 for idempotent writes
Build	EAS Build (cloud), `preview` profile makes an installable APK

12.1 Source layout

```
Milltrack/
  app.json, eas.json, package.json
  docs/                     this documentation (+ screenshots)
  src/
    app/                    screens (every file is a route)
      _layout.tsx           providers, auth gate, AppState -> focusManager
      index.tsx             role-based redirect
      login.tsx
      (gate)/  (godown)/  (supervisor)/  (packaging)/  (admin)/
    components/
      gate/                 GateButton, GateInput, SelectField, ConfirmDialog, Toast, StatusBadge
      godown/ supervisor/ packaging/ admin/   role-specific components
      DateStepper.tsx
    constants/gateTheme.ts  the single design system (colours, radii, shadows)
    lib/
      api.ts                every backend call, one place
      types.ts              API types
      AuthContext.tsx       session
      OfflineQueueContext.tsx, offlineQueue.ts
      supervisor.ts cone.ts unload.ts dates.ts format.ts   domain helpers
```

12.2 Role routing

`app/index.tsx` reads the stored role after login and redirects: `GM_MD_ADMIN` to Admin, `GODOWN_MANAGER_PKG` to Packaging, `SUPERVISOR` to Supervisor, any other godown role to Godown, anything else to the Gate. Each group's layout also guards its screens. The server enforces permissions again on every request, so hiding a screen is a convenience, not the security boundary.

12.3 API conventions

- All calls go through `apiFetch` in `lib/api.ts`: adds the bearer token, parses JSON, and turns failures into readable messages.
- Errors arrive as `{"detail": ...}`; the message is shown as is, including 409 conflicts.
- Network failure shows "Cannot reach the server" with the address and the underlying error to aid diagnosis.
- Decimal quantities (kg, hours, percentages) are strings in JSON and are never converted to floating point for storage.
- Writes that must not duplicate (gate entries, bales, DOFFs, shift rows, bags) carry a client-generated id, so retrying after a timeout is safe.

12.4 Design system

All colours, radii and shadows are tokens in `constants/gateTheme.ts`. Components use the tokens, not hard-coded values, so the whole app can be re-themed in one file. A single light theme is used. Main colour: **#0369A1**.

12.5 Offline behaviour

Only the Gate entry has an offline queue (vehicles are registered at gates with poor signal). Other screens need a connection and say so. React Query keeps the last loaded data on screen while it refreshes.

12.6 Build and release

1. Set `EXPO_PUBLIC_API_BASE_URL` in `.env` (compiled into the app at build time).
2. Android package name and icons are in `app.json`. Plain HTTP is permitted through the `expo-build-properties` plugin (`usesCleartextTraffic`) because the server currently has no TLS certificate.
3. `npx eas-cli@latest build --platform android --profile preview` creates the APK. The signing keystore is held by EAS.
4. Run `npx tsc --noEmit` and `npx expo lint` before building. The lock file must be in sync (`npm ci` is used by the cloud build).

> **RECOMMENDED HARDENING**
>
> Serve the API over HTTPS and then remove the cleartext allowance. Until then, traffic including passwords travels unencrypted and should only be used on trusted networks.

12.7 Endpoint reference

Area	Endpoints
Auth / config	`POST /auth/login`, `GET /config/lookups`
Gate	`POST /gate/uploads/presign`, `/presign-bill`, `POST/GET /gate/entries`, `POST /gate/entries/{id}/exit`
Godown	`GET /godown/unloading-queue`, `POST /godown/bales`, `GET/PATCH /godown/entries/{id}/bales[/tare]`, `POST /godown/bales/{id}/void|print`, `POST /godown/entries/{id}/finish-unloading`, `GET /godown/stock-bales`, `GET /godown/inventory`, `GET/POST /godown/mixing-lots`, `.../{id}/scan|waste|complete`
Supervisor	`GET /supervisor/current-shift`, `GET/POST/DELETE /supervisor/machines`, `POST /supervisor/shift-logs/{process}`, `POST /supervisor/shift-submissions`, `POST /supervisor/doff`, `GET /supervisor/doff/last|total/{machine}`
Packaging	`POST /packaging/records`, `GET /packaging/records/available`, `GET /packaging/loose-stock`, `GET /packaging/loading-vehicles`, dispatch entries/lines/bags, `PUT/GET /packaging/paper-log`, `POST /packaging/finalize-day`
Admin	`GET/POST /admin/users`, `PATCH /admin/users/{id}`, `POST /admin/users/{id}/revoke`, `POST /admin/pallis`, `PUT /admin/mixing-recipe`, `GET /admin/analytics/*`, `GET /admin/trace/{code}`, `GET /admin/reports`

Two endpoints, `/packaging/finalize-day` and `/admin/reports`, were built from the backend team's written specification and were not yet in `ApiDocumentation.json` when this was written.

13. Known limitations and open items

Item	Detail
Label printing	The server print call exists; the on-phone printer hook is a placeholder until the label printer is chosen.
Dispatch history	A dispatch older than yesterday is not found by the list screen.
Finish loading	There is no "finish loading" endpoint yet; a dispatch is closed by the vehicle exiting.
Dispatch QR	A dispatch QR code is not shown.
Packed bags	There is no void or edit for a packed bag.
Run / idle time	Entered as decimal hours. If the mill's paper uses H.MM, validation must change. Awaiting confirmation.
Paper sheet questions	"Miss Coloured", one unlabelled column, and the other bag weights are waiting on the client's answers.
Pallis list	Shows active pallis only, because no admin list endpoint exists.
Security	Plain-HTTP API at present (see Section 12.6).
Testing	Screens were exercised in a browser against simulated server responses and on a real Android phone for sign-in. A full on-device pass with live data across all roles is recommended before go-live.

14. Glossary

Term	Meaning
Bale	A compressed bundle of raw cotton. Each gets a unique QR code.
Gross / tare / net	Gross is the weighed total; tare is the weight of wrapping and straps; net = gross minus tare.
Party	A client of the mill. A bale and its yarn stay linked to its party.
Palli	A client plus a bobbin colour. Packaging packs bags against a Palli.
Mixing lot	A batch of bales (and own waste) blended in the blow room to a recipe.
Own waste	The mill's own recycled waste added back into a mix.
Blow room, Card, Draw Frame, O/E	The production stages: cotton opening and mixing, carding, drawing, and open-end spinning (making yarn).
DOFF	Removing a full package of yarn from an O/E machine. Loose yarn produced this way goes to loose stock.
Shift A / B	Shift A starts 7 am, Shift B starts 7 pm.
Loose stock	Yarn cones not yet packed into a bag; not yet owned by any client.
Cone type	How many cones make a bag and what each cone weighs.
Count	The fineness of yarn (for example 20s, 30s).
Idempotent	Safe to repeat: sending the same request twice has the same effect as once.
KRA / KPI	Key result area (what a role is accountable for) / key performance indicator (how it is measured).